Jood A. Al Aswad

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Professional Appointments

Postdoctoral Fellow, Virginia Polytechnic Institute and State University 2025–present
Laboratory of Dr. Pedro Monarrez · Department of Geosciences.

Education

Ph.D. Geological Sciences: Quantitative Paleobiology, Stanford University 2025
M.S. Geological Sciences: Stratigraphy & Petrophysics, Cornell University 2019
B.S. Earth Science: Concentration in Geology, George Mason University, top student in program 2016

Grants and Fellowships

Thomas D. and Janice H. Barrow Graduate Fellowship (~$56,000) 2024–2025

Graduate Fellowship, King Abdullah Scholarship Program: Stanford (~$205,000) 2019–2022

Graduate Fellowship, King Abdullah Scholarship Program: Cornell (~$150,000) 2017–2019

Donovan Family Scholarship: Cornell ($2,000) 2018

Full Ride Scholarship, King Abdullah Scholarship Program 2011–2016

Undergraduate Student Research Grant, URSP, George Mason University (~$1,500) 2015

Scientific Publications

7. **Al Aswad, J.,** Bazzi, M., Penn, J., Monarrez, P.M., Deutsch, C., and Payne, J. Coupling of climate change and extinction drive taxonomic homogenization in deep time. Proceedings of the Royal Society B. In review

6. **Al Aswad, J.,** Hautmann, M., Singh, P., Ferrill, N.L., Al-Ramadan, K., Lehrmann, D.J., Morsilli, M.D., Koeshidayatullah, A.I., and Payne, J.L. Bivalves from the Lower Triassic of central Saudi Arabia: Systematics and calibration of Permo-Triassic boundary position within the Khartam Member of the Khuff Formation. In prep.

5. Bazzi, M., Lloyd, W.L., Ebersole, J., Sternes, P., **Al Aswad, J.,** and Payne, J. Extinction threatens to cause morphological and ecological homogenization in sharks. Science Advances. DOI: 10.1126/sciadv.aea0278 2025

4. **Al Aswad, J.,** Penn, J., Bazzi, M., Monarrez, P.M., Deutsch, C., and Payne, J. Physiology and climate change explain unusually high similarity across marine communities after end-Permian mass extinction. Science Advances. DOI: 10.1126/sciadv.adr4199 2025

3. Payne, J.L., **Al Aswad, J.,** Deutsch, C., Monarrez, P.M., Penn, J.L., and Singh, P. Selectivity of mass extinctions: Patterns, processes, and future directions. Cambridge Prisms: Extinction. DOI: 10.1017/ext.2023.10 2023

2. Tester, J., Beyers, S., Gustafson, J.O., Jordan, T.E., Smith, J.D., **Al Aswad, J.A.,** et al. District geothermal heating using EGS technology to meet carbon neutrality goals: A case study of Earth Source Heat for the Cornell University campus. Proceedings World Geothermal Congress. 2020

1. Gustafson, J.O., Smith, J.D., Beyers, S.M., **Al Aswad, J.A.,** Jordan, T.E., and Tester, J.W. Earth Source Heat: Feasibility of deep direct use of geothermal energy on the Cornell campus. GRC Transactions, Vol. 4. 2018

Honors

Research and Innovation Postdoctoral Fellow 2026
Virginia Tech

Winifred Goldring Award (Outstanding Paleontology Ph.D. Student) 2025
Association for Women Geoscientists & The Paleontological Society

Centennial Teaching Assistant Award 2025
Stanford University

Harriet Benson Fellowship Award for Excellent Research 2022
Stanford University

Best Scientific Video: Storytelling 2020
Stanford BioX

Certificate of Highest GPA in Program 2016
George Mason University

Outstanding Senior in Earth Science Award 2016
George Mason University

Leadership and Service

Creator & Committee Leader, DEEP Paleontology Awards 2022–2023
The Paleontological Society. Founded the Diversity, Equity, and Excellence in Paleontology (DEEP) Awards from the ground up to recognize outstanding scholars from underrepresented groups.

Access, Belonging & Community Liaison 2024
Stanford Earth

Committee Member 2021–2025
The Paleontological Society

Student Representative, Paleontological Society Council 2021–2023

Outreach Funding

Al Aswad, J. Funds for DEI Grants Program, The Paleontological Society (~$10,000). Secured funding and founded the PaleoSociety DEI Grants Program to support underrepresented researchers in paleontology. 2023

Al Aswad, J. Grant for DEEP Paleontology Awards, The Paleontological Society (~$12,000). Secured funding and created the DEEP Awards from the ground up. 2022

Workshops and Educational Training

Introduction to Earth System Modeling Workshop 2026
UC Riverside · Instructor: Dr. Andy Ridgwell

Best Practices in Teaching and Mentoring Workshop 2026
Virginia Tech Center for Teaching and Learning

Course Design Institute 2025
Stanford University Center for Teaching and Learning

Preparing Future Professors Program 2023
Mentor: Dr. K. Allison Meezan, Foothill College

TEACH Symposium 2022
Stanford University

Marine Invertebrate Zoology 2021
University of Washington, Friday Harbor Laboratories · Grade: 99/100

Field Camp 2015
South Dakota School of Mines and Technology · Grade: A

Teaching Experience

Teaching Assistant: EPS 169/269 The Sixth Mass Extinction, Stanford University 2023 x2, 2024
Devoted one full quarter exclusively to course development with Dr. Jonathan Payne: built the syllabus from scratch, designed all activities, problem sets, and lecture materials for a new course on mass extinctions. Served as head TA for two subsequent teaching quarters, personally delivering multiple standalone lectures each term. Involved in all three iterations of the course from inception through delivery.

Teaching Assistant: EPS 136/236 Macroevolution, Stanford University Spring 2024
Supported lecture and discussion sections for upper-division macroevolution course covering speciation, phylogenetics, and deep-time diversity patterns.

Teaching Assistant: GEOLSCI 4 Coevolution of Life and Earth, Stanford University Autumn 2022

Guest Lecturer: Physical Geography (Climate Change), Foothill College Winter 2023
Taught Week 7 on climate change; designed hands-on activities including geologic time visualization and climograph exercises.

Module Instructor: Introduction to Paleobiology, Friday Harbor Laboratories / Univ. of Washington Summer 2021
Introduced paleobiology to undergraduate and graduate students; incorporated active-learning technology (TopHat).

Student Mentoring

Pokémon-Inspired Evolution Museum Event 2026–present
Virginia Tech Museum of Geosciences. Co-created and mentored graduate students in creating interactive public event teaching children about evolution and biodiversity.

Undergraduate & high school research mentor and mentor assistant 2022
Stanford SURGE, Bio-X, and Stanford Earth Young Investigators programs: Victor Trujillo, Lucy Helms, Edward Huang, Kelly Tung, McKenna Sanders, Sakeena Saber.

Peer Advisor 2019–2020
Bay Area Graduate Pathways in STEM

Mentoring undergraduate and graduate students 2025–present
Virginia Tech

Presentations

Invited Presentations

"The Great Dulling: climate, thermal physiology, and mass extinctions explain when and why global communities homogenize in deep time." To be presented at Geological Society of America, Colorado. 2026

"Synergy of climate change and biodiversity loss drive global taxonomic homogenization in deep time." Departmental Seminar Series, George Mason University. 2026

"Examining the compositional similarity of marine invertebrate ecosystems after the Big Five mass extinction events." Fossil Coffee Series, UC Berkeley. 2024

"Taxonomic homogenization of marine ecosystems after the end-Permian mass extinction." Oxford University Biogeography Workshop, Oxford, UK. 2024

"Physiology of marine invertebrates explains broadened biogeography after end-Permian extinction." University of Lausanne, Switzerland. 2023

Selected Conference Presentations. ** denotes student mentee*

Trujillo, V.*, Helms, L.*, Al Aswad, J.A., Monarrez, P., and Payne, J.L. "Biogeographic stability of benthic marine mollusks across the end-Cretaceous mass extinction." GSA Annual Meeting, Denver. 2022

Tung, K.*, Anderson, M.*, Saber, S.*, Monarrez, P.M., Al Aswad, J.A., and Payne, J.L. "Comparing the latitudinal ranges of genera of Mollusca and Arthropoda before and after the end-Permian mass extinction." AGU Fall Meeting, Washington D.C. 2022

Al Aswad, J.A. "Porosity, permeability, and paleotopographic analysis: A geological interpretation of geothermal reservoir quality at Cornell University." GSA Annual Meeting, Indianapolis. 2018

Al Aswad, J.A. "The harmonic analysis of geophysical phenomena recorded by borehole strainmeters." OSCAR Students as Scholars Symposium, George Mason University.

Professional and Community Service

Committee Member, Early Career Research Symposium 2026–present
International Fossil Coral and Reef Society

Session Co-Chair, "Linking Biodiversity Loss to Environmental Stressors" 2025
Geological Society of America

Session Co-Chair, "Quantifying Organism-Environment Interactions" 2024
Geological Society of America

Session Co-Chair, "Past and Future Organism-Environment Interactions" 2023
Geological Society of America

Program Coordinator, Science Teaching Through Art 2021–2022
Stanford University

Coordinator, Geological Sciences Graduate Seminar Series 2019–2022
Stanford University

Committee Member 2020
North California Paleontological Conference

President, AAPG Student Chapter 2017–2019
Cornell University

Reviewer for: *Science; Gondwana Research; Science Advances; Palaeogeography, Palaeoclimatology, Palaeoecology*

Technical Skills & Competencies

Statistical modeling & data science
R/RStudio (intermediate), MATLAB and Python (beginner)

Earth system modeling
cGENIE (climate dynamics, biogeochemical cycles)

Fossil identification
Marine invertebrate paleobiology (bivalves)

Field skills
Paleontological and stratigraphic field work, geological mapping

Design
Adobe Creative Suite, digital art

Languages
English (fluent), Arabic (conversational)